Reg No: 22BAL1266

Name:

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

Continuous Assessment Test-II, Winter Semester, March 2025

Programme	**B.Tech**	CAT	**2**
Course Title	**Microeconomics**	Course Code	**BHUM103L**
		Slot	**C1+TC1**
Faculty	**Shalina Susan Mathew**	Class ID	**CH2024250501365**
Duration	**90 minutes**	Max. Marks	**50**

Answer all the questions

Sl. No	Question	Marks
1.	(a) Explain economies and diseconomies of scale, with relevant diagram. (b) Refer to the following table. The input price for both capital and labour are Rs. 12 per unit (i.e., w = r = Rs. 12 per unit). **Output** / **Labour** / **Capital** 1 / 1 / 1 3 / 2 / 2 6 / 4 / 4 8 / 8 / 8 Based on the table, discuss whether the firm experiences economies/diseconomies of scale, and if so, over what range of production of output. Show relevant calculations.	10
2.	For each of the following situations, determine whether the firm is concerned with a short-run or a long-run production decision. Explain briefly in each case. (a) An automobile firm manager decides to add an extra six-hour shift each day to keep the assembly line running 24 hours per day. (b) The ONGC decides to start drilling in the complex and ultra-deep waters of the Cauvery basin. (c) After studying a report on the predicted increase in the life expectancy in the state of Tamil Nadu, a group of doctors decides to establish a new geriatric hospital. **(…CTND… PTO)**	10

(d) The COO of IKEA India decides not to build another distribution centre in Gurugram, instead choose to extend hours of operation by 10 percent at each of the existing centres.

3. (a) Explain elasticity of demand. By elaborating on the sign (+/-), describe the cross-price elasticity of demand, with at least one example.

(b) For each of the following scenarios, decide whether you agree or disagree and explain your answer.

(i). Suppose the price elasticity of demand for public hospital service is -0.15, and it does not vary much between the rural and urban patients. The government is starting to charge the urban patients 50 percent more. The Department of Health hopes that the changes would help it increase the revenue.

(ii). Destination weddings have become more popular these days. During the preceding three years, the market has been competitive; as a result, price has fallen by some 20 percent. If the price elasticity of demand was -1.6, vendors would lose revenues altogether as a result of the price decline. (Assume there was no shift in the demand curve in the last three years.)

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4. The following table shows the alternative combinations of capital (K) and labour (L) required to produce 50 and 100 units of output.

	Q = 50		Q = 100	
	K	L	K	L
A	1	8	2	10
B	2	5	3	6
C	3	3	4	4
D	5	2	6	3
E	8	1	10	2

(a) Define isoquant. Based on the table above, draw an isoquant map representing the different output levels. *(While labelling the graph, denote Labour on x-axis).*

(b) Define MRTS. For the output level 100, between points E and D, what is the MRTS of capital for labour?

(c) 'Isoquants are convex to the origin, and two isoquants cannot intersect.' Explain whether you agree/disagree with this statement.

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Total Marks **[50]**